

Finding the Gap, Reviewing Literature, and Positioning Your ML Paper

2-page handout for CS / ML students writing conference papers (NeurIPS, ICML, ICLR, AAAI, ACL, CVPR)

1. What counts as a research gap in ML/CS?

A research gap is a **meaningful unresolved issue** in prior work. In ML papers, strong gaps often come from unrealistic assumptions, weak evaluation, missing comparisons, poor scalability, or missing theoretical explanation.

Gap type	ML-style question	Example
Problem gap	What part of the technical problem is still not solved?	Existing MARL methods do not handle non-stationary mixed traffic well.
Assumption gap	What unrealistic assumption do prior papers make?	Prior work assumes centralized training signals or full observability.
Evaluation gap	What is not tested rigorously enough?	Methods are evaluated only on toy benchmarks, not realistic urban networks.
Comparison gap	Which baseline, ablation, or setting is missing?	Papers compare against weak baselines and omit decentralized alternatives.
Scalability / robustness gap	What breaks under realistic scale, noise, or shift?	Performance degrades when agent count or graph size grows.

Weak gap: “Deep RL is important, and more work is needed.”

Better gap: “Existing decentralized MARL approaches for route choice are mainly evaluated in small synthetic settings and rarely model mixed autonomy, leaving their stability and scalability in realistic urban networks unclear.”

2. How to find a gap from papers

For each paper, extract six items:

Item	Question to ask while reading
Task	What problem do they solve?
Setting	What data, simulator, benchmark, or environment do they use?
Method	What method family do they use?
Assumptions	What information, supervision, or structure do they assume?
Evidence	What experiments, ablations, or theory support the claim?
Limitation	What is missing, weak, unrealistic, or untested?

Practical reading prompt: “This paper shows that ... However, it assumes / does not test / does not explain ...”

Gap formula: “Prior work has shown *[what is known]*, but still relies on *[limiting assumption / weak evaluation]*, making it unclear whether *[important unresolved question]*.”

3. Literature review in ML: synthesize, do not enumerate

A weak background section lists papers one by one. A strong one groups papers into families, compares assumptions and strengths, and shows what remains missing.

Useful organizing schemes: by method family (value-based, actor-critic, communication), by challenge (non-stationarity, credit assignment, scalability), or by benchmark realism (toy tasks, synthetic graphs, realistic simulators).

Review paragraph template: “Existing work on *[topic]* can be grouped into *[family 1]*, *[family 2]*, and *[family 3]*. Family 1 is strong at *[x]* but weak at *[y]*. Family 2 addresses *[x]* yet assumes *[y]*. Family 3 improves *[x]* but has not been tested on *[y]*. Therefore, *[gap]*.”

4. Building your contribution

Your contribution is not just “we apply method X.” It should state **what is new**, **relative to what**, **why it matters**, and **what evidence supports it**.

Level	Question	Example
Method	What did you build or do?	A decentralized communication-aware policy learner.
Evidence	What did you demonstrate?	It improves stability and travel time on realistic traffic networks.
Significance	Why should the field care?	It narrows the gap between toy MARL benchmarks and deployment-relevant routing.

Contribution formula: “We propose / present / show *[what]*, which differs from prior work by *[key distinction]* and enables / demonstrates *[why it matters]*.”

5. CARS for ML introductions

Move	Purpose	ML-style example
1. Establish territory	Show the problem matters.	Multi-agent learning in non-stationary environments remains a central challenge in RL.
2. Establish niche	Show what is missing.	Existing approaches often assume centralized information or evaluate only simplified benchmarks.
3. Occupy the niche	Present your paper.	We propose and evaluate a decentralized framework for mixed-traffic route choice on realistic urban networks.

5-sentence ML intro skeleton: Problem → Limitation → Method / study → Evidence → Contribution.

6. Positioning table: compare your paper to the closest prior work

Study	Task / setting	Method family	Main limitation	Relation to our work
Paper A	Cooperative routing on small graphs	QMIX / VDN	No mixed traffic; toy scale	We test larger, realistic networks.
Paper B	Decentralized MARL	IPPO / MAPPO	No route-choice realism	We apply decentralized learning to realistic routing.
Paper C	Communication MARL	Message passing	Not evaluated in transportation	We test communication in routing.
This work	Mixed-traffic route choice	Your method	—	—

7. Fast checklist before writing

- I can name the 3–5 closest papers to mine.
- I know their task, assumptions, and evaluation setting.
- My gap is specific, meaningful, and not generic.
- My literature review is organized by ideas, not author names.
- My contribution directly addresses the identified gap.
- I can explain in one sentence how my paper differs from prior work.
- My introduction follows problem → limitation → method → evidence → contribution.

Mini fill-in exercise

Gap: Prior work on *[topic]* has shown *[known result]*, but it still assumes / misses *[limitation]*, making it unclear *[open question]*.

Contribution: We *[propose / benchmark / analyze / show]* *[your contribution]*, which differs from prior work by *[key distinction]* and demonstrates *[main finding / value]*.